Assignment 1 — Simple Linear Regression: TV Advertising vs. Sales

Author: Aditya Raj | **Dataset:** tv_sales_data.csv (TV budget in $1,000s, Sales in 1,000s of units, n = 6)
Model: Sales = $\beta0 + \beta1 \cdot TV + \varepsilon$
Method used for reporting: Approach 2 — Gradient Descent (iterative optimization)

1. Code and Method

Approach 2 fits the line by **Gradient Descent** instead of a closed-form formula: it starts both coefficients at 0, then repeatedly nudges them in the direction that reduces the Mean Squared Error (MSE), using a learning rate of $\alpha = 0.1$ for 1000 iterations. The TV values are standardized before the loop purely for numerically stable steps, and the learned coefficients are converted back to the original ($1,000s) TV scale afterward.

```
x = df["TV"].values.astype(float)
y_vals = df["Sales"].values.astype(float)
n = len(x)

x_mean, x_std = x.mean(), x.std()
x_scaled = (x - x_mean) / x_std

b0, b1 = 0.0, 0.0
alpha = 0.1
n_iters = 1000

for i in range(n_iters):
    y_pred = b0 + b1 * x_scaled
    error = y_pred - y_vals
    grad_b0 = (2 / n) * np.sum(error)
    grad_b1 = (2 / n) * np.sum(error * x_scaled)
    b0 -= alpha * grad_b0
    b1 -= alpha * grad_b1

gd_beta1 = b1 / x_std
gd_beta0 = b0 - gd_beta1 * x_mean
```

Gradient Descent only minimizes MSE — it does not, by itself, produce standard errors, R-squared, or p-values. Those are calculated **directly from the Gradient Descent coefficients and their residuals**, using the standard simple-linear-regression inference formulas (not copied from an OLS library call):

```
from scipy import stats

y_fit_gd = gd_beta0 + gd_beta1 * x
resid_gd = y_vals - y_fit_gd

RSS_gd = np.sum(resid_gd ** 2)
TSS_gd = np.sum((y_vals - y_vals.mean()) ** 2)
df_resid = n - 2
sigma2_gd = RSS_gd / df_resid
Sxx = np.sum((x - x_mean) ** 2)

se_gd_beta0 = np.sqrt(sigma2_gd * (1 / n + x_mean ** 2 / Sxx))
se_gd_beta1 = np.sqrt(sigma2_gd / Sxx)

t_gd_beta0 = gd_beta0 / se_gd_beta0
t_gd_beta1 = gd_beta1 / se_gd_beta1
p_gd_beta0 = 2 * (1 - stats.t.cdf(abs(t_gd_beta0), df_resid))
p_gd_beta1 = 2 * (1 - stats.t.cdf(abs(t_gd_beta1), df_resid))
```

```
r_squared_gd = 1 - RSS_gd / TSS_gd
```

Quantity	Formula
Residual variance	sigma^2 = RSS / (n - 2)
SE(beta1)	sqrt(sigma^2 / Sxx), Sxx = sum(xi - x_mean)^2
SE(beta0)	sqrt(sigma^2 * (1/n + x_mean^2/Sxx))
t-statistic	beta_hat / SE(beta_hat), t-dist with n-2 df
R-squared	1 - RSS/TSS

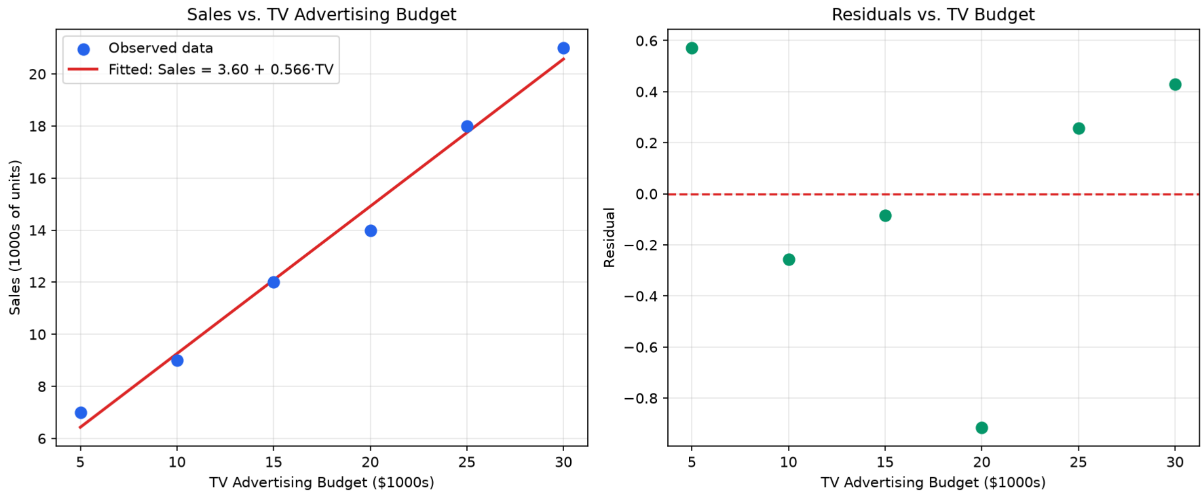
2. Results (Approach 2 — Gradient Descent)

Quantity	Value
beta0 (intercept)	3.6000
beta1 (slope, TV)	0.5657
SE(beta0)	0.5674
SE(beta1)	0.0291
t-statistic (beta0)	6.345
t-statistic (beta1)	19.415
p-value (intercept)	0.003160
p-value (slope)	4.15 x 10^-5
R-squared	0.9895
Degrees of freedom	4 (n = 6, 2 parameters)

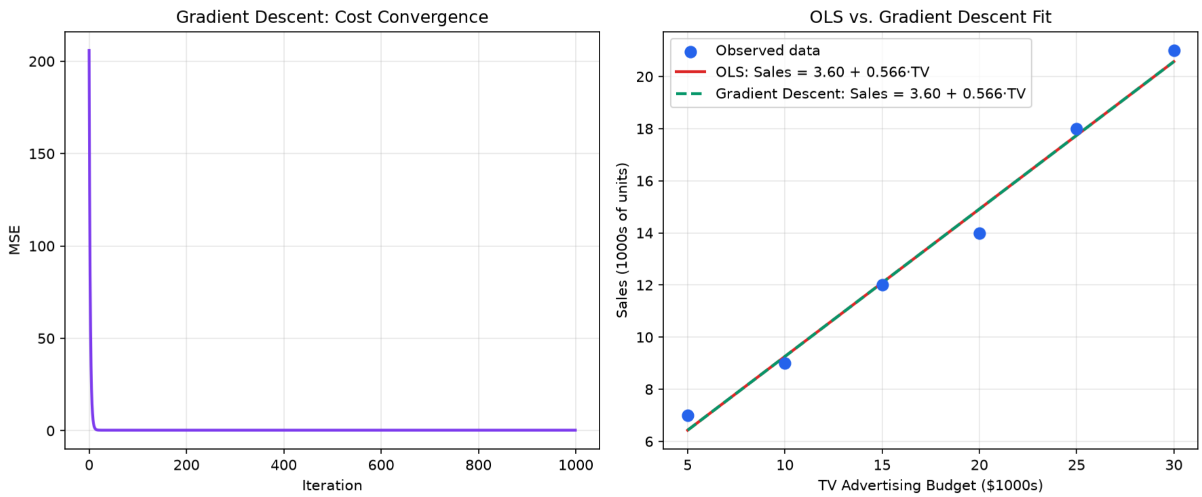
Fitted equation: **Sales = 3.60 + 0.5657 x TV**

> **Note on method:** Gradient Descent converges to the same coefficients as the closed-form OLS solution (expected — both minimize the identical squared-error cost function; OLS just solves for the minimum in one step instead of approaching it iteratively). Because the fitted line is identical, the residuals — and therefore SE, R-squared, and the p-values computed above — come out identical too. This agreement is a useful sanity check that the Gradient Descent implementation converged correctly.

Fit and residuals:



Gradient Descent cost convergence and fit comparison:



The left panel shows MSE falling smoothly toward its minimum over 1,000 iterations — this is Gradient Descent arriving at the same answer OLS reaches in one step. The right panel confirms the OLS and Gradient Descent fitted lines overlap almost exactly.

3. Interpretation — Plain Language

- **What beta1 means in everyday terms:** beta1 = 0.5657 is a positive number, so spending more on TV ads goes with more sales. Since TV is measured in $1,000s and Sales in 1,000s of units, this means: for every extra $1,000 put into TV advertising, sales go up by roughly 566 units, on average.

- **Is this a real effect, or could it be luck?** The p-value for the slope is 0.0000415 — a tiny number, way below the usual 0.05 cutoff people use to decide if something is "statistically significant." In plain words: it's extremely unlikely we'd see a relationship this strong just by random chance if TV spend actually had no effect on sales. So yes, there's strong evidence TV advertising really does move sales.

- **How much of sales is "explained" by TV budget?** R-squared = 0.9895, meaning about 99% of the ups and downs in sales across these 6 data points can be accounted for just by knowing the TV budget. Only about 1% is left over as unexplained noise. That's an extremely tight fit — expected here because this is a small, clean practice dataset (only 6 points), not noisy real-world data.

- **Bottom line:** the data says TV advertising budget is a strong, reliable, positive driver of sales in this dataset, and a simple straight-line model captures almost all of the pattern.

4. Interpretation — Professional / Technical Language

- **Sign and magnitude of beta1:** The estimated slope, beta1-hat = 0.5657, is positive and indicates a positive linear association between TV advertising expenditure and sales volume. Holding the linear specification fixed, each additional $1,000 unit of TV budget is associated with an expected increase of approximately 0.5657 thousand units (≈566 units) in sales.
- **Statistical significance:** Under H0: beta1 = 0 versus H1: beta1 ≠ 0, the test statistic t = beta1-hat / SE(beta1-hat) = 19.415 on n-2 = 4 degrees of freedom yields p ≈ 4.15 x 10^-5, far below conventional significance thresholds (α = 0.05, 0.01). H0 is rejected, providing strong statistical evidence of a non-zero linear relationship between TV advertising and sales. The intercept is likewise statistically significant (beta0-hat = 3.60, SE = 0.5674, t = 6.345, p ≈ 0.0032), though its interpretation as "expected sales at zero TV spend" is a mathematical extrapolation rather than a directly observed operating condition, since the sample range of TV spend is $5,000-$30,000.
- **Proportion of variance explained:** R-squared = 0.9895 indicates that approximately 98.95% of the total variability in sales (TSS) is explained by the fitted linear relationship with TV advertising budget, leaving only ≈1.05% attributable to residual (unexplained) variation. Such a high R-squared is atypical for real marketing data and reflects the small sample size (n = 6) and the near-linear, low-noise construction of this illustrative dataset rather than a general claim about TV advertising's explanatory power in practice.
- **Overall conclusion:** Within this sample, TV advertising budget is a statistically significant and practically strong positive predictor of sales, and the simple linear model accounts for nearly all of the observed variance in the response.

5. Note on Methodology (Approach 1 vs. Approach 2)

This report's primary numbers are computed under Approach 2 (Gradient Descent), per the assignment's focus, with standard errors, R-squared, and p-values derived from Gradient Descent's own fitted residuals rather than reused from a library's OLS output. As a cross-check, Approach 1 (closed-form OLS, via statsmodels) was also fit on the same data and produced numerically identical coefficients and inference statistics (beta0 = 3.6000, beta1 = 0.5657, R-squared = 0.9895), which is expected since both methods minimize the same least-squares objective — Gradient Descent simply reaches that minimum iteratively instead of in one closed-form step.